

EXERCISE NO. 5

Interfaces

1. Start by making a normal Java-7-style application with these features:
 - An interface called RegularPolygon with two abstract methods: getNumSides and getSideLength
 - A class EquilateralTriangle that implements the interface, has getNumSides return 3 and getSideLength return an instance variable that is set by the constructor.
 - A class Square that implements the interface, has getNumSides return 4 and getSideLength return an instance variable that is set by the constructor.
2. Add a static totalSides method, that given a RegularPolygon[], returns the sum of the number of sides of all the elements.
3. Add two default methods: • getPerimeter ($n * \text{length}$, where n is the number of sides) • getInteriorAngle ($(n-2)\pi/n$ in radians).

Code:

```
import java.util.*;

interface RegularPolygon{
    // Abstract methods
    int getNumSides();
    double getSideLength();

    // Static method
    static int totalSides(RegularPolygon [] polygons){
        int total = 0;

        for(RegularPolygon polygon: polygons){
            total = total + polygon.getNumSides();
        }
        return total;
    }

    // Default methods
    default double getPerimeter(){
        return getNumSides() * getSideLength();
    }

    default double getInteriorAngle(){
        return (getNumSides() - 2) * Math.PI / getNumSides();
    }
}
```

```

// Equilateral Triangle
class EquilateralTriangle implements RegularPolygon{
    private double sideLength;

    // Constructor
    EquilateralTriangle(double sideLength){
        this.sideLength = sideLength;
    }

    @Override
    public int getNumSides(){
        return 3;
    }

    @Override
    public double getSideLength(){
        return sideLength;
    }
}

class Square implements RegularPolygon{
    private double sideLength;

    Square(double sideLength){
        this.sideLength = sideLength;
    }

    @Override
    public int getNumSides(){
        return 4;
    }

    @Override
    public double getSideLength(){
        return sideLength;
    }
}

public class Main {
    public static void main(String[] args) {
        EquilateralTriangle triangle = new EquilateralTriangle(5);
        Square square = new Square(10);
        System.out.println("Triangle sides: " + triangle.getNumSides());
        System.out.println("Triangle side length: " + triangle.getSideLength());
        System.out.println("Triangle perimeter: " + triangle.getPerimeter());
        System.out.println("Triangle interior angle: " + triangle.getInteriorAngle());
    }
}

```

```
System.out.println();

System.out.println("Square sides: " + square.getNumSides());
System.out.println("Square side length: " + square.getSideLength());
System.out.println("Square perimeter: " + square.getPerimeter());
System.out.println("Square interior angle: " + square.getInteriorAngle());
System.out.println();
// Array of RegularPolygon objects
RegularPolygon[] polygons = { triangle, square };
System.out.println("Total sides: " + RegularPolygon.totalSides(polygons));
}
}
```